



Web -Based Remote Monitoring, Electricity Cost Modeling, and Decision Support System Powered by Artificial Intelligence



The Task

Using the [PGCB Hourly Generation Dataset \(Bangladesh\)](#) provided with this assignment, build forecasting models and present the historical data together with your forecasts in an interactive dashboard. *[We care more about clear, working end-to-end work than a perfect accuracy score. Even if you cannot finish all of the works, you are encouraged to submit.]*

Part A: Machine Learning & Deep Learning

1. Do brief EDA and preprocessing (handle missing values, resample, add time features like hour/day/lags).
2. Pick one target to forecast (e.g., electricity demand or generation) and state your forecast horizon.
3. Build at least one classical ML model (e.g., Random Forest / XGBoost or any other model you like) and one deep learning model (LSTM / GRU or any other model you like).
4. Use a time-based train/test split (no shuffling) and evaluate with MAE, RMSE, and MAPE.
5. Plot actual vs. predicted and briefly say which model did the best.

Part B: Interactive Dashboard (Web Dashboard if Possible)

1. A time-series graph/chart of the historical data.
2. The forecasted future values overlaid (distinct color; confidence band if possible).
3. A small summary panel (e.g., latest value, peak, average, model error etc) and one interactive control (date range or model selector).

Deliverables

1. A short report: approach, model results table and dashboard screenshots.
2. A 2 minute demo video (unlisted YouTube / Drive link). (You can use any language for the demonstration)

You can use any type of AI for the entire work. There is no restriction.

Dataset Link:

[https://archive.ics.uci.edu/dataset/1175/pgcb+hourly+generation+dataset+\(bangladesh\)](https://archive.ics.uci.edu/dataset/1175/pgcb+hourly+generation+dataset+(bangladesh))

Submission Link:

https://docs.google.com/forms/d/e/1FAIpQLSeQdhpG3m8sYdy9yJJtsYzaH5TJGdJObOmf13ot1Bg_MFmIVA/viewform?usp=header

Deadline: 5 August 2026 at 10pm